Scores are capped at 100 points.

1. (10 points) A language is **P**-*complete* if it is in **P** and every language in **P** is logspace reducible to it. Show that if L is **P**-complete, then $L \in$ uniform **NC** iff **P** = uniform **NC**.

2. (30 points) Define $\mathbf{TC}^0$ as the class of languages that can be computed by constant-depth, polynomial-size circuits with unbounded fan-in *threshold gates*. A threshold gate $T_{\boldsymbol{w},\theta}$ outputs 1 if the weighted sum of the inputs is at least some threshold, and outputs 0 otherwise. i.e.

$$T_{\boldsymbol{w},\theta}(x_1,\ldots,x_n) = \mathbb{I}\left\{\sum_{i=1}^{n} w_i x_i \geq \theta\right\}, w_i, \theta \in \mathbb{Z}.$$

Note that the length of binary representation of w_i and θ is at most poly(n). $\mathbf{TC}^0$ holds practical significance in the field of neural networks. For example, a single forward pass of a fixed-precision, finite-depth Transformer can be simulated by $\mathbf{TC}^0$.

The majority function $Maj_n : \{0,1\}^n \to \{0,1\}$: outputs 1 if the number of 1s in the input is at least $n/2$; outputs 0 otherwise.

   (a) (10 points) Show that Maj cannot be computed in $\mathbf{AC}^0$ by reducing XOR to it.

   (b) (10 points) Show that threshold gates can be simulated by $\mathbf{NC}^1$ circuits.

   (c) (10 points) Show that $\mathbf{AC}^0 \subsetneq \mathbf{TC}^0 \subseteq \mathbf{NC}^1$.

3. (30 points) We know that PTMs have 2 transition functions δ_0 and δ_1. Now fixed some $p \in (0,1)$. At each step, we choose δ_0 with probability p and δ_1 with probability $1-p$. Then define $\mathbf{BPP'}$ as the class of languages that can be decided by such PTMs with error probability at most 1/3.

   (a) (10 points) Show that $\mathbf{BPP} \subseteq \mathbf{BPP'}$.

   (b) (10 points) Show that $\mathbf{BPP} = \mathbf{BPP'}$ for each $p \in \mathbb{Q}$.

   (c) (10 points) Show that $\mathbf{BPP} \subsetneq \mathbf{BPP'}$ for some p.

4. (25 points) Language A reduces to language B under a *randomized polynomial time reduction*, denoted $A \leq_r B$, if there is a PTM M s.t. for every $x \in \{0,1\}^*, \Pr[B(M(x)) = A(x)] \geq 2/3$.

   Define $\mathbf{BP} \cdot \mathbf{NP} = \{L : L \leq_r \text{3SAT}\}$.

   (a) (15 points) A nondeterministic circuit C has two inputs x, y. We say that C accepts x iff there exists y such that $C(x,y) = 1$. The size of the circuit is measured as a function of $|x|$. Let $\mathbf{NP}/_{\mathbf{poly}}$ be the languages that are decided by polynomial size nondeterministic circuits. Show that $\mathbf{BP} \cdot \mathbf{NP} \subseteq \mathbf{NP}/_{\mathbf{poly}}$.

   (b) (10 points) Show that if $\overline{\text{3SAT}} \in \mathbf{BP} \cdot \mathbf{NP}$, then **PH** collapses to $\mathbf{\Sigma}_3^p$.

5. (25 points) In this problem, we will discuss the power of SAT oracles.

   (a) (15 points) Show that $\mathbf{BPP} \subseteq \mathbf{ZPP}^{\text{SAT}}$.

   (b) (10 points) If $\mathbf{NP} \subseteq \mathbf{BPP}$, show that $\mathbf{P}^{\text{SAT}} \subseteq \mathbf{BPP} = \mathbf{ZPP}^{\text{SAT}}$.